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Original Research Article

What influences farmers' intentions to conserve on-farm biodiversity? An application of the theory of planned behavior in fars province, Iran

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ABSTRACT

Earth's well-being depends mainly on the conservation of ecosystem goods and services. However, biodiversity is extensively threatened in agricultural landscapes. Since farmers' decisions and practices are the main determinant factors of on-farm biodiversity conservation, this study was conducted to clarify the underlying constructs of farmers' intention to preserve biodiversity. The Theory of Planned Behavior was applied to investigate the intention of a sample of 274 Iranian farmers towards on-farm biodiversity conservation. In order to increase the predictive power of this theory, the new constructs of knowledge, moral norm and perceived threats of intensified agriculture were added to the model. The results indicated that farmers have a moderate level of knowledge about the value and strategies of biodiversity conservation. Also, they have a favorable attitude toward preserving biodiversity. However, they do not perceive enough control to engage in biodiversity conservation practices. Furthermore, the findings revealed that attitude, perceived threats of intensified agriculture, knowledge, social norms and moral norms are the most relevant factors explaining the farmers' intentions toward biodiversity conservation. In general, the results presented evidence about effectiveness of the Theory of Planned Behavior and the extended model of planned behavior on assessing the farmers' intention to preserve biodiversity. Also, the findings reinforced the available evidence regarding the significance of moral norm and knowledge in development of environmental intention. Accordingly, a number of considerations about improving farmers' intention to preserve on-farm biodiversity are recommended.

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1. Introduction

Nearly 40% of the Earth's surface is under use for food and fiber production. As a result of such intense agricultural activity, reducing wild flora and fauna species have become one of the major concerns for environmentalists, policy makers and the scientific communities (Park, 2015). Although intensive agriculture in pure sense has not really happened in developing countries, some agriculture related activities in developing countries have made essential changes in the chemical composition of the soil and water resources and, also, the flora and fauna of the farm ecosystems. These activities include

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uncontrolled use of chemical inputs on farms, intensive tillage, introduction of new hybrid crop species, aggression of agricultural lands from their true borders and rapid changes in agricultural land use (Firbank et al., 2008; Scherr and McNeely, 2008; Greiner, 2015; Coromaldi et al., 2015). Changes in the agricultural activities and farm management styles have led to biological oversimplification of farms' environment over the past century (Greiner, 2015; Park, 2015; Deng et al., 2016).

Given the concerns raised about the environmental impacts of agriculture, various governments have considered biodiversity conservation at the farm level. For instance, the Australian government has developed a project entitled "flagship natural resource management initiative" with the budget of $2 billion. This project provides the farmers with information about land and biodiversity management practices on farm with the aim of improving soil properties and contributing to environment protection (Barson et al., 2012). Also, in Hong Kong, the government has formulated the Biodiversity Strategy and Action Plan (BSAP) for 2016-2021 to step up biodiversity conservation and support sustainable development. Through setting out a framework of actions, BSAP will facilitate enhancement of conservation measures, mainstream biodiversity, advance knowledge and help the community to involve in biodiversity conservation (Hong Kong environment Bureau, 2016). Likewise, over the past 15 years, the Iran' government has planned many interventions such as reducing over-consumption of chemical pesticides and fertilizers, improving irrigation water management and facilitating adoption of environmentally sound technologies to mitigate the negative effects on the agricultural landscapes and preserve biodiversity. However, many top down interventions have failed to conserve biodiversity and biodiversity loss is still the great concern of environmental supporters in Iran (United Nations Development Programme, 2016).

Currently, about 16.5 million hectares of the Iran' lands are used for agricultural production (Statistical Centre of Iran, 2014). This country is one of the main countries of the Middle East in terms of needing biodiversity conservation (Farashi et al., 2017). Although a large part of Iran is dry, its specific and varied geography, climate, topography and geological structure have led to a unique biodiversity in the country. "Over 8200 species of plants, 197 species of mammals, 535 species of birds, 227 species of reptiles, 21 species of amphibians, 160 species of freshwater fishes and 710 species of marine fishes have been recorded in Iran" (Iranian Department of Environment, 2016). Some medicinal and aromatic species of Iran have a high commercial value throughout the world (The Fifth National Report to the Convention on Biological Diversity, 2015). It is while biodiversity in Iran has faced many threats. So that many valuable species have been degraded as a result of irrational use of soil and water resources over the past few decades. Meantime, the annual soil erosion has exceeded 1.5 billion tons and deforestation has increased. Also, more than 50% of the wildlife sanctuaries and over 70% of the protected areas require a long time to recover due to the over extraction of vegetation (The Forth National Report to the Convention on Biological Diversity, 2010).

Since farmers have direct interaction with natural ecosystem at the farm level (Jackson et al., 2010), they can help to conserve the genetic resources on their agricultural lands and the surrounding areas (Guillem et al., 2015). However, the farmers' intention to preserve biodiversity on the farm depends on a large number of factors (Herzon and Mikk, 2007; Ahnström et al., 2008). In order to design effective interventions that can promote biodiversity on farm, clarification of these determinant factors is imperative. Therefore, this study was conducted to investigate the drivers of farmers' intention to preserve on-farm biodiversity.

Different models and theories have been introduced for recognizing and predicting human intention or behavior, e.g. Health Belief Model by Rosenstock (1974), Norm Activation Model by Schwartz (1977), Trans-theoretical Model by Prochaska and Di Clemente (1982), Theory of Reasoned Action (TRA) by Ajzen and Fishbein (1980) and Theory of Planned Behavior (TPB) by Ajzen (1991). This study applies TPB as a basis for identification of the main drivers of the farmers' intention to conserve on-farm biodiversity. TPB is a powerful tool to simplify complex decision contexts (e.g. biodiversity conservation) and make them measureable through three predictor constructs, i.e. perceived behavioral control, social norms and attitudes. Many relevant determinants that have been introduced in the other theories are background factors simply mediated through these three intention antecedents. Moreover, TPB has been found to be well supported by empirical evidences, and many studies (e.g. Fielding et al., 2008; Greaves et al., 2013; Yazdanpanah and Forouzani, 2015; Meijer et al., 2015; Botetzagias et al., 2015; de Leeuw et al., 2015; Poškus, 2015; Halder et al., 2016; Gao et al., 2017; Zhang et al., 2017) have confirmed effectiveness of this theory for explaining environmental intentions and behaviors. However, TPB has been criticized for mere emphasis on logic in behavior formation and, also, the vacancy of some emotional (Gao et al., 2017; de Leeuw et al., 2015), mental and perceptual constructs (Rosenstock, 1974; McFarlane and Boxall, 2003; Fabrigar et al., 2006; Pieniak et al., 2010; Raymond and Schneider, 2014; de Leeuw et al., 2015; Janmaimool, 2017; Liu et al., 2017), which have important roles in predicting and changing environment-related behaviors. Given the emphasis placed on openness of TPB for including the important constructs that are able to increase the predictive power of this model (Ajzen, 1991), in this study, some moral and perceptual variables are added to TPB in order to investigate their impact on the farmers' intention to preserve biodiversity. To the best of our knowledge, the farmers' willingness to preserve on-farm biodiversity has not been studied in Iran. Therefore, the findings of this study will provide new insight about farmers' intention and behavior towards on-farm biodiversity conservation in order to guide formulation of biodiversity conservation policies and programs.

1.1. The extended model of planned behavior

TPB is a significant socio-cognitive model for predicting individuals' behavior and intentions (Yazdanpanah and Forouzani, 2015). Based on this theory, the intention to perform a behavior is predicted by attitude, subjective norm and perceived behavior control (PBC; Ajzen, 1991). PBC is defined as the people's perception about having control over a particular behavior

and their ability to engage in that behavior (Ajzen and Fishbein, 1980). Attitude is the individual's desirable or undesirable emotions towards a behavior while subjective norm is the people's perception of the expectations and community's support for doing that behavior (Ajzen, 1991). Based on TPB, the people, who have a positive attitude towards a behavior, imagine they will be supported by the community if they do the behavior, recognize that they are able to get involved in the behavior and, therefore, they have a strong intention to perform the behavior (Fielding et al., 2008).

TPB is one of the most widely used theories for anticipating pro-environmental behaviors. For example, Meijer et al. (2015) used this theory to investigate the tree planting behavior of smallholder farmers in Malawi. They concluded that attitude has the most significant effect on the tree planting behavior. Also, Halder et al. (2016) applied TPB to explore the intention of Finnish and Indian students to use bioenergy and found that attitude has the strongest effect on the students' intentions to use bioenergy. In their study, subjective norm was shown to have a significant effect on intention whereas perceived behavioral control was found to have a negligible effect. Using TPB, de Leeuw et al. (2015) assessed the key beliefs underlying pro-environmental behavior among high-school students in Luxembourg. Their findings revealed an excellent fit for the standard TPB model. Also, adopting an extended model of TPB, Yazdanpanah and Forouzani (2015) predicted Iranian students' intention to purchase organic foods. They found that the students' attitude was the main predictor of their intention to purchase organic foods while the perceived behavioral control and subjective norms were not significant predictors of the intention (see also Van Thanh and Yapwattanaphun, 2015; Chen, 2017; Botetzagias et al., 2015; Greaves et al., 2013).

Although the effectiveness of TPB in predicting intention has been confirmed in various fields, Ajzen (1991) stated that the theory is essentially open to add additional explanatory constructs if it can be proved that the new constructs can help to explain a significant portion of the variance of intention and behavior. Therefore, a large number of the environmental studies that apply TPB attempt to incorporate additional constructs into this theory in order to increase its predictive power (e.g. Hoeksma et al., 2017; Gao et al., 2017; Maichum et al., 2016; Shin and Hancer, 2016; Yazdanpanah and Forouzani, 2015; Botetzagias et al., 2015). Among the various constructs that have been added to TPB, moral norm has a special position (Botetzagias et al., 2015) and different studies have incorporated this construct into TPB to enhance its predictive power (e.g. Liu et al., 2017; Zhang et al., 2017; Yazdanpanah and Forouzani, 2015; Raymond and Schneider, 2014). Moral norm is a moral commitment that people feel for doing a specific behavior (Bamberg and Möser, 2007). Since farmers' decision about conservation of farm biodiversity is influenced by logical and emotional factors, in the present study, moral norm was added to TPB as an additional construct for studying farmers' intention to conserve biodiversity (Fig. 1).

Literature review also indicated that knowledge can influence intention by giving people the ability to formulate strong arguments in supporting their intentions (Fabrigar et al., 2005; McFarlane and Boxall, 2003; Pieniak et al., 2010; Malek-Saeidi et al., 2012). Therefore, biodiversity knowledge was concerned in the expanded model of planned behavior, too (Fig. 1). Biodiversity knowledge includes farmers' knowledge regarding the concept of biodiversity, its value for human and the environment, and the strategies for on-farm biodiversity conservation.

Moreover, based on the Health Belief Model (HBM), one of the main determinants of behavior in a specified situation is whether individuals perceive a threat about that situation or not (Rosenstock, 1974). This is especially important when the perceived threat tends to be less serious than an actual threat (Mullan et al., 2013). Various studies have considered perceived threat for investigation of pro-environmental behaviors (e.g. Janmaimool, 2017; Marquit, 2008; Johnson and Scicchitano, 2000). Stern (1992) has proposed that threat perception about environmental problems is an element of "anthropocentric

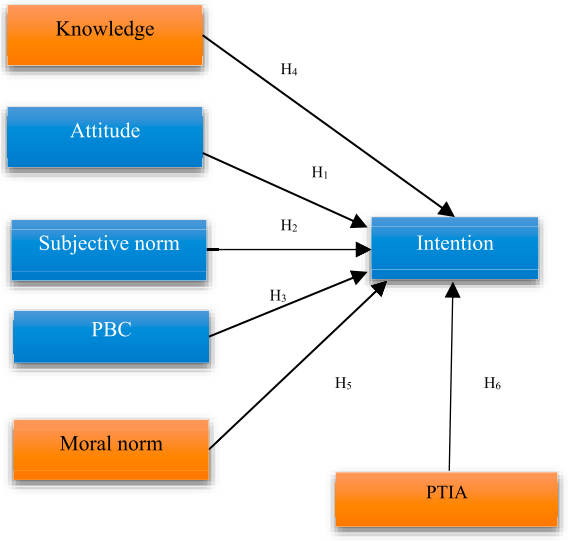


Fig. 1. The extended model of the theory of planned behavior.

value orientation". Based on this viewpoint, when people feel that the environment is in a danger, they would worry about it and feel that they are responsible for maintaining environmental quality. It seems that farmers' intention to preserve on-farm biodiversity is influenced by their perception about the destructive effects of agriculture on biodiversity. Therefore, perceived threat of intensified agriculture (PTIA) to biodiversity was added to TPB to enhance its predicting power (Fig. 1).

As indicated in Fig. 1, it is hypothesized that:

H1. Attitude would positively influence farmers' intention to conserve biodiversity;

H2. Subjective norm would positively affect intention;

H3. PBC would positively impact intention;

H4. Knowledge would positively predict intention;

H5. Moral norm would positively influence intention; and

H6. PTIA would positively affect intention.

2. Materials and methods

2.1. Geographical context of the study area

Fars is located in southwestern Iran (Fig. 2), at the 50°30′ to 55°38′ longitude and the 27°3′ to 31°42′ N latitude. Fars, with an area of 122,608 km^2 and a population over 4,800,000 is the third largest province in the country (Keshavarz et al., 2016). Agricultural sector is the main source of income for the Fars rural communities. While agriculture and animal husbandry are the mainstay of its rural economy, other activities such as handcrafting have provided off-farm incomes for rural families. Also, this province, with its numerous cultural (e.g. Saadi and Hafiz tomb), historical (e.g. Persepolis), natural (internationally renowned lakes such as Kaftar and Bakhtegan) and religious (e.g. Shah-e-Cheragh shrine) attractions, is one of the major tourism' centers of Iran. Owing to its unique position, eco-tourism has been considered as an alternative source of income, too.

The Fars province has a wide variety of climate from warm and dry in the southern and eastern parts to cold and humid in the north. The temperature of Fars varies from −10 °C in winter to 45 °C in summer (Hojjati and Boustani, 2010). The annual rainfall in various areas of Fars is 100–800 mm (Nafarzadegan et al., 2012). Fars is one of the leading agricultural areas in the

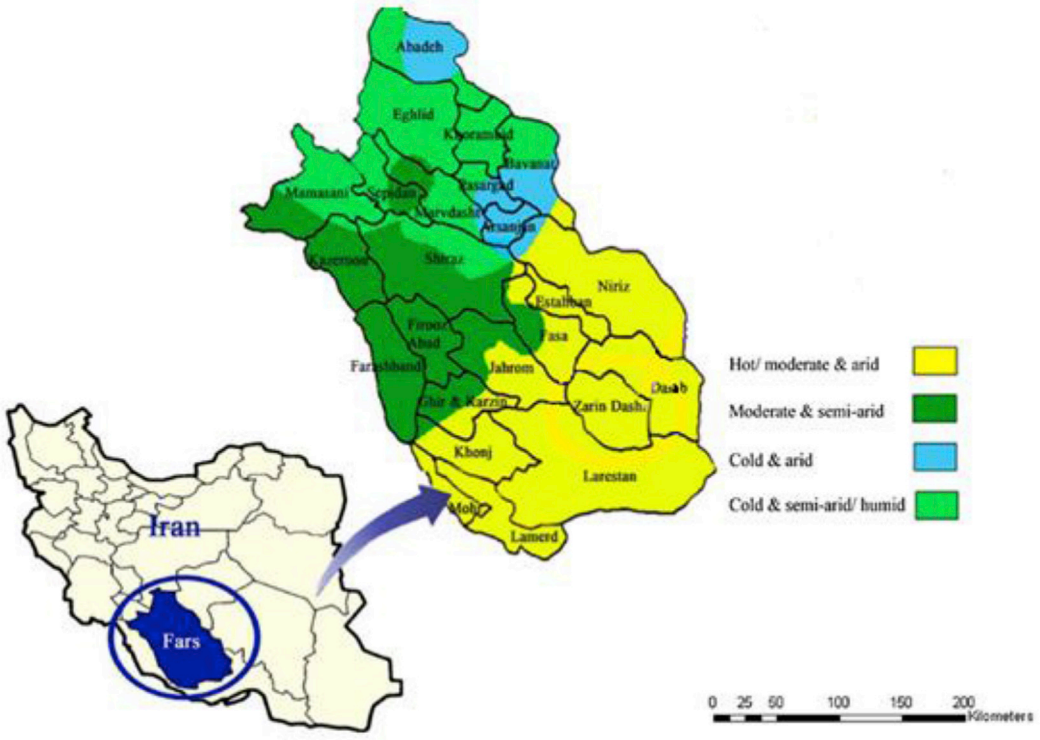


Fig. 2. The study area.

country (Keshavarz and Karami, 2014). For many years, this province has ranked first in production of wheat (Nafarzadegan et al., 2012). Furthermore, Fars in known for production of barley, rice, maize and sugar beet. Cultivation of rice, cotton, oilseed, potato, onion, tomato, pomegranate, spinach and forage crops are common in this province, too. Fig. 3 indicates some photos of the farming landscapes in the study area.

Specific topography and climatic conditions along with other special environmental conditions, such as being placed on the slopes of the Zagros Mountains, have led to huge diversity in the ecosystems and habitats of this province (Bayat and Hashemi Hosseini, 2010). However, decades of unplanned development, population growth and agricultural intensification have threatened its fragile ecosystems and rich biodiversity (United Nations Development Programme, 2016). Due to the fact that Fars is still an important area in terms of biodiversity (existence of some valuable and rare plants species in this area; Dolatkhai et al., 2010), this province was selected as the study area.

2.2. Participants

The target population of the study was the farmers of the Fars province. The sample size (n = 274) was specified using the formula proposed by Scheaffer et al. (1979). The multistage stratified random sampling method was applied for selecting the sample farmers. In the first step, according to the Fars climate map, the province was divided into four climatic stratums that included: (1) cold and dry; (2) warm/moderate and dry; (3) cold and semi-humid; and (4) moderate and semi-dry areas. In the second step, a number of counties were chosen proportionally from each stratum. In the third step, a number of sub-counties were selected from each county. In the fourth step, two villages were chosen from each sub-county. Finally, in proportion to the population of each village, a number of farmers were chosen. Sample selection was performed randomly in all outlined steps.

2.3. Data collection techniques

The data for this cross-sectional survey study were collected in 2017. Farmers were interviewed individually through a structured questionnaire. The interviews lasted about 25–30 min and the response rate was 100% due to the farmers' interest in discussing the studied issue and their presence at home in off season for agriculture during the February–March interval. The researchers tried to gather the data carefully and accurately.

The questionnaire was divided into two parts: (i) the socio-economic attributes of the study sample (including age, farm ownership, and education level); and (ii) predictors of the farmers' intention to conserve biodiversity, i.e. attitude, subjective norm, perceived behavior control, knowledge, moral norm and perceived threat of intensified agriculture (see Table 1 for description of variables and their measurements). Face validity of the scales applied to the study was confirmed by a panel of agricultural and environmental experts. In order to measure reliability of the instrument, a pilot study was conducted on 30 farmers in the Firuzabad County, a region out of the study area. The Cronbachs' alpha coefficients for the Likert type scales were estimated between 0.66 and 0.80 (Table 1).



Fig. 3. Agricultural farm landscapes in the study area.

Table1
Variables, definition, measurement and Cronbachs' alpha coefficients.

Variables	Definition and measurement	Items	Cronbachs' alpha coefficient
Intention	Farmer's tendency to conserve on-farm biodiversity; ordinal items ranging from 1 (strongly disagree) to 5 (strongly agree).	(1) I am willing to work with the environmental advocacy groups. (2) I am willing to reduce the use of insecticides to help prevent insects from being destroyed. (3) I am ready to avoid consumption of chemical herbicides in order to maintain biodiversity. (4) I would like to conserve biodiversity in the next few years by planting various plant species on my farm.	0.80
Attitude	Farmer's desirable or undesirable emotions towards preserving on-farm biodiversity; ordinal items ranging from 1 (strongly disagree) to 5 (strongly agree).	(1) Conserving biodiversity is more important than increasing agricultural production. (2) The killing of wild animals just due to their attack on agricultural crops is not correct. (3) Protection of plant and animal species is essential for production of food, medicine and other crops. (4) To maintain biodiversity on the farm, it is essential to remove chemical insecticides.	0.68
Subjective norm	Farmer's perception of the community's support for preserving biodiversity; ordinal items ranging from 1 (very low) to 5 (very high).	(1) My family members have a positive view about protection of plant and animal species. (2) Most people who matter to me are keen on protecting plant and animal species. (3) If more farmers support biodiversity, I will also try more to support it. (4) As the government supports biodiversity, I will try hard to conserve it.	0.72
PBC	Farmer's perception about their ability to engage in the biodiversity conservation activities; ordinal items ranging from 1 (strongly disagree) to 5 (strongly agree).	(1) I believe that I am able to increase the population of useful insects on my farm. (2) I have enough facilities to preserve on-farm biodiversity. (3) If I want to, I can increase the population of native plants on my farm. (4) I have everything that is required to raise livestock along with crop activities. (5) I have enough facilities to control weeds manually.	0.78
Knowledge	Farmer's knowledge about the concept, value and strategies of on-farm biodiversity conservation; ordinal items ranging from 1 (very low) to 5 (very high).	(1) I am aware about the impact of crop rotation on biodiversity conservation. (2) I am aware from the importance of biodiversity conservation on the farm. (3) I have enough information about strengthening the natural enemies of pests to prevent pest infection. (4) I know how to protect wild plant species inside and around the farm for biodiversity conservation. (5) I am aware of the importance of restoring the crop residues in the soil for biodiversity conservation. (6) I know the importance of using organic matter and compost for biodiversity conservation.	0.66
Moral norm	The amount of moral obligation that farmers feel towards on-farm biodiversity conservation; ordinal items ranging from 1 (very low) to 5 (very high).	(1) I feel responsible for protecting indigenous plants on the farm. (2) I consider the prevention of wildlife hunting as a moral obligation. (3) Reduction of insecticides' use is a moral responsibility. (4) I am responsible for learning more about how to conserve biodiversity on farm. (5) I am responsible to encourage other farmers to reduce the use of chemical fertilizers on farm.	0.76

Table1 (*continued*)

Variables	Definition and measurement	Items	Cronbachs' alpha coefficient
		(6) I consider protection of forests, pastures and protected areas as a moral responsibility.	
PTIA	Farmer's perception about the negative impacts of intensified agriculture on biodiversity at the farm level; ordinal items ranging from 1 (very low) to 5 (very high).	(1) Cultivation of modified seeds in farms has led to the disappearance of native plant species. (2) Application of insecticides has adverse effects on the insects' population. (3) Consumption of chemical fertilizers and pesticides in agriculture threatens many animals and native birds. (4) Expansion of agricultural land has led to the elimination of many native plant species in the region. (5) Increasing the chemical content of drainage water is a threat to animal species in the region.	0.69

2.4. Construction of the extended TPB model and statistical analysis

In order to determine the causal directions between the constructs of the extended model of planned behavior (EMPB), i.e. attitude, subjective norm, perceived behavior control, knowledge, moral norm and perceived threat of intensified agriculture, and their mutual intervening effects, the Bayesian network was applied (Wu, 2010). The Bayesian network as a data mining technique demonstrates the probabilistic relationships between multiple constructs. The WEKA software was used to represent the conditional probability distributions as a directed acyclic graph (DAG). Also, Tree augmented Naïve Bayes (TAN) was considered to search the Bayesian network space. TAN treats the class attribute (i.e. intention) as the supreme parent node. Following these analyses, the causal relationship diagram was attained (Fig. 4).

After determining the probabilistic relationships between the constructs, confirmatory factor analysis was used to make sure that the analysis results are not influenced by any estimation biases. The results indicated that all EMPB constructs can certainly be distinguished empirically and reflect different attributes. For examination of the research hypotheses, the standard TPB and EMPB were tested through conducting structural equation modeling (SEM) using the AMOS software version 23. SEM as a multivariate statistical analysis method combines factor analysis and multiple regression analysis to estimate the structural relations between the measured variables and the latent constructs, which are extracted from the manifest variables (Vinodh and Joy, 2012).

Moreover, to describe the farmers' profile, descriptive statistics was used through the SPSS software version 23. Also, to investigate the relations between the specified constructs, Pearson correlation was applied.

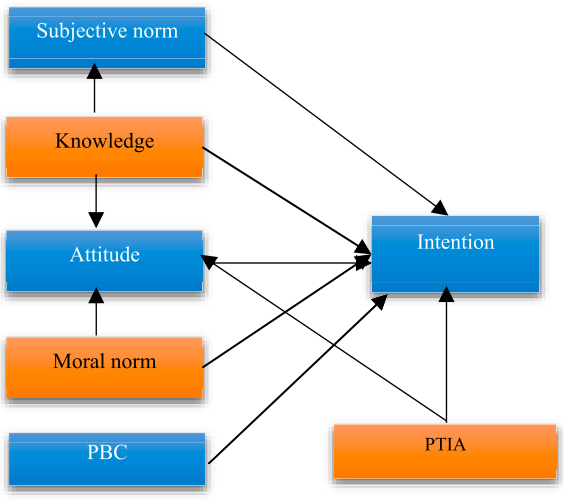


Fig. 4. Causal relationships between the EMPB constructs.

3. Results and discussion

3.1. Participants' profile

The participants were aged from 25 to 70 and their average age was about 47 years (SD = 10.96). The average years of the farmers' education was 5.84 (SD = 5.02). This result is consistent with the findings of Aref (2011) and Maleksaeidi et al. (2016). The low level of education among Iranian farmers is due to their limited access to educational facilities and resources in rural communities. Also, most farmers owned small scale farms and smallholder farming was the main type of agricultural production system in the study area. The average farm size of the participants was 5.50 ha (SD = 4.23). The mean values of intention (14.08), attitude (14.03), subjective norm (14.70), moral norm (23.78) and PTIA (18.32) were relatively high. The mean score of knowledge was moderate (16.32) while the mean value of PBC was low (9.82). These findings suggested that the studied farmers had a moderate level of knowledge about the value of conserving biodiversity and the required strategies. Also, they showed a favorable attitude toward preserving biodiversity and, in order to protect it, they felt morally committed. However, they did not perceive enough control to engage in biodiversity conservation practices. This may be due to the fact that they do not have the required tools and skills to improve biodiversity on the farm and conserve it.

3.2. Relationships between variables

Table 2 shows the results of Pearson correlation test between the variables. As shown in Table 2, there is a significant correlation between intention towards biodiversity conservation with most variables including attitude, subjective norm, knowledge, moral norm and PTIA. However, there is no significant relationship between intention and PBC.

3.3. Determinants of farmers' intention to preserve biodiversity conservation

In order to determine the drivers of farmers' intention to preserve biodiversity, SEM was used. In this regard, two steps were considered. First, TPB and EMPB were evaluated through confirmatory factor analysis. This step demonstrates that "whether the theoretical structural model fits the observed variables" (Deng et al., 2016). After that, direct, indirect and total effects of the variables were calculated.

3.3.1. Model evaluation

Some indicators were applied to verify appropriateness of the studied models for analyzing the survey data and estimate fitness of the TPB and EMPB models. The fit indices for the studied models are reported in Table 3. As shown in this table, all indicators confirmed goodness of fit for the two models. For instance, the Chi-square ratio to degree of freedom (χ^2/df) for TPB and EMPB were estimated as 3.09 and 3.59, which are less than the acceptable value of 5 (Meyers et al., 2006). Also, the values of the Goodness of Fit Index (GFI) as "a measure of fit between the hypothesized model and the observed covariance matrix", Comparative fit index (CFI) as "the model fit by examining the discrepancy between the data and the hypothesized model" and Bentler-Bonett normed fit index (NFI) as "the discrepancy between the chi-squared value of the hypothesized model and the chi-squared value of the null model" were higher than 0.9 for the TPB and EMPB models, which meant acceptable model fitness.

3.3.2. Drivers of the farmers' intention to preserve biodiversity based on the TPB model

As shown in Fig. 5, the basic model of TPB explained 53% of the variance in the farmers' intention towards on-farm biodiversity conservation. Based on this model, attitude indicates the maximum influence on intention ($\beta = 0.59$; $p < 0.001$). This finding is similar to the results of Meijer et al. (2015) and Yazdanpanah and Forouzani (2015). Confirming the first research hypothesis (H_1), this result implies that the farmers who had a positive attitude towards biodiversity conservation were more eager to preserve biodiversity on their farms. Also the findings illustrated that subjective norm has significantly affected the farmers' intention to preserve biodiversity ($\beta = 0.40$; $p < 0.001$). The significant influence of subjective norm on intention confirms the second research hypothesis (H2) and implies that the farmers' decision-making process and their intention towards biodiversity conservation is influenced mainly by their friends, family and other

Table 2
Correlation coefficients between variables.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
(1) Intention	1					
(2) Attitude	0.64**	1				
(3) Subjective norm	0.52**	0.58**	1			
(4) PBC	−0.14	−0.16	−0.11	1		
(5) Knowledge	0.14*	0.41**	0.29**	0.07	1	
(6) Moral norm	0.46**	0.48**	0.49**	−0.07	0.26**	1
(7) PTIA	0.49**	0.54**	0.55**	−0.12*	0.24**	0.49**

The significance of test: *P < 0.05 and **P < 0.01.

Table 3
Goodness of fit indices for TPB and EMPB.

Fit index	Recommended value[a]	Estimate value for	
		TPB	EMPB
Satorra-Bentler scaled Chi-Square (χ^2)	-	422	688
Normed Chi-Square (χ^2/df)	<5	3.09	3.59
Goodness of-fit index (GFI)	≥0.9	0.91	0.92
Parsimonious goodness-of-fit index (PGFI)	>0.5	0.59	0.61
Comparative fit index (CFI)	≥0.9	0.99	1.00
Parsimonious comparative fit index (PCFI)	>0.5	0.63	0.65
Bentler-Bonett normed fit index (NFI)	≥0.9	0.90	0.99
Parsimonious normed fit index (PNFI)	>0.5	0.62	0.64
Root-mean square error of approximation (RMSEA)	<0.10	0.061	0.001

[a] Adopted from Meyers et al. (2006).

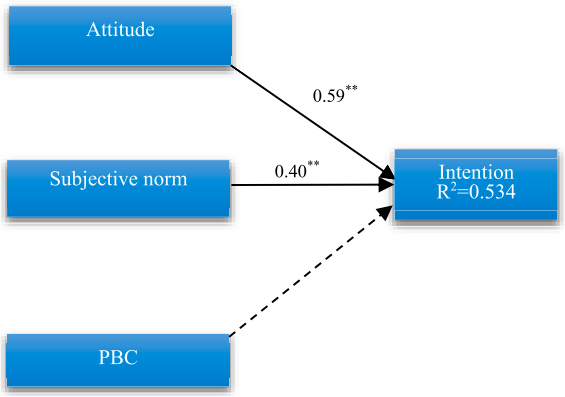


Fig. 5. Standard path coefficients for TPB.

important people in the community. Therefore, in order to promote biodiversity conservation, technical interventions such as biological control of weeds and pests, improvement of water use efficiency and cultivation of cover crops should be accompanied with effective social supports.

Interestingly, direct association between PBC and intention was not statistically significant (Fig. 5). This means that, contrary to the basic assumption of TPB, i.e. the third hypothesis (H_3), PBC is not a predictor of intention in this study.

3.3.3. Drivers of the farmers' intention to preserve biodiversity based on EMPB

Based on Fig. 6, EMPB has a higher predictive power relative to standard TPB. While standard TPB illustrated 53% of the variance of intention, EMPB explained 67% of the variance of conservation intention among Iranian farmers. This finding showed that incorporation of the new constructs to standard TPB (i.e. the knowledge, moral norm and PTIA constructs) has improved the explanatory power of the TPB model for predicting intention to preserve biodiversity. Fig. 6 reveals that attitude ($\beta = 0.55$; $p < 0.001$) and subjective norm ($\beta = 0.20$; $p < 0.001$) were significantly associated with intention to preserve biodiversity and, like the TPB model, attitude exerts the maximum effect on intention. This means that attitude is the main driver of the farmers' intention towards biodiversity conservation. Yazdanpanah and Forouzani (2015) similarly found out that attitude and subjective norm have significant effects on the intention of purchasing organic productions. Moreover, the findings illustrated that the farmers' intention was significantly influenced by PTIA as a non-TPB attribute ($\beta = 0.33$; $p < 0.001$). This finding, which confirms the sixth hypothesis of the research (H_6), implies that the farmers, who perceived the negative contribution of intensified agriculture (e.g. application of chemical inputs) on farm-level biodiversity, more likely intended to conserve their farms and the surrounding environment. Similar studies conducted by Flaherty et al. (2019), Janmaimool (2017) and Marquit (2008) demonstrated that the people, who perceive the destructive impacts of their behaviors on the environment, are more inclined to adopt protection strategies. In fact, as long as the farmers are not aware of the negative effects of their activities on biodiversity, they are not expected to take steps to improve their management activities and protect biodiversity. Also, the findings indicated that knowledge, as another non-TPB construct, has significantly affected the farmers' intention towards biodiversity conservation ($\beta = 0.13$; $p < 0.05$). This finding supports the fourth hypothesis of the research (H_4) and suggests that farmers' knowledge about the concept, value and strategies of on-farm biodiversity conservation has positive effects on their attitude and such farmers are more interested in conducting conservation practices. Direct effect of knowledge on attitude has been confirmed in several studies (e.g. Bakhtiari et al., 2014;

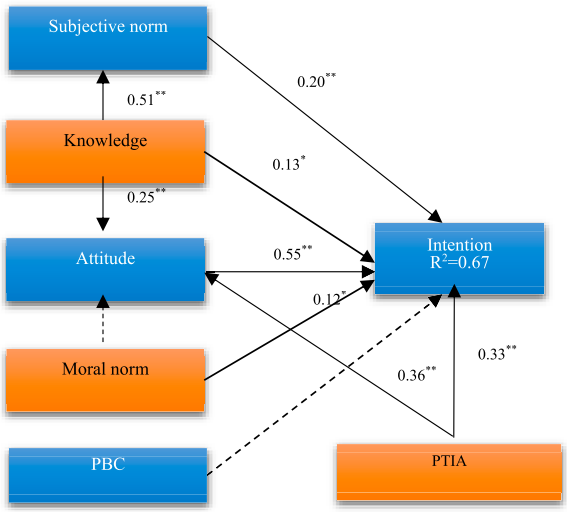


Fig. 6. Standard path coefficients for EMPB.

Malek-Saeidi et al., 2012; Niankara and Zoungrana, 2018; Wu, 2010). Furthermore, moral norm, as another non-TPB construct, was significantly associated with intention ($\beta = 0.12$; $p < 0.05$). This finding approves the fifth hypothesis of the research (H_5) and points out to the fact that the farmers, who felt more moral commitment to the environment, perceive that adherence to the nature and adoption of environmentally sound strategies is essential to avoid the adverse impacts of farm management efforts on the environment. This finding is similar to the results of Liu et al. (2017); Yazdanpanah and Forouzani (2015) and Zhang et al. (2017). They asserted that individual behavior is not only influenced by logic and cost-benefit calculations, and found out that emotions and moral obligations also have determinant roles in developing pro-environmental behaviors. Similar to the standard model of TPB (Fig. 5), the association between PBC and intention (the third hypothesis of the research) was not significant using EMPB. This is mainly because the agriculture sector is heavily dependent to the government's financial and technical supports. Most Iranian farmers receive many supports from the government to ensure food supply. Therefore, they believe that the government should invest in farm-level conservation of biodiversity, too. Moreover, insignificant effect of PBC on intention can be the result of inadequate access to information and farm management facilities and, also, lack of skill to apply conservation strategies on farm. Insignificant influence of PBC on intention rejects the initial assumption of the present study and is contrary to the findings of some previous studies (e.g. Zhang et al., 2017; Gao et al., 2017). However, this result is consistent with the findings of Yazdanpanah and Forouzani (2015).

The findings also revealed that attitude and subjective norm mediated the effects of knowledge and PTIA on intention to preserve biodiversity. It means that the farmers, who perceived more threat from the intensified agricultural systems had a positive attitude towards biodiversity preservation and desired to protect on-farm biodiversity more. Moreover, the farmers' attitude and subjective norms were significantly influenced by their knowledge about the value of biodiversity conservation. This implies that being more aware about the value of biodiversity can change the farmers' attitude towards environment conservation and give them the power to more effectively defend from their attitude in relation to biodiversity conservation. Also, environmental knowledge stimulates the farmers' estimation of their ability in supporting on-farm biodiversity conservation.

Overall, attitude ($\beta = 0.55$; $p < 0.001$), PTIA ($\beta = 0.52$; $p < 0.001$), knowledge ($= 0.36$; $p < 0.001$), subjective norms ($\beta = 0.20$; $p < 0.01$), and moral norms ($\beta = 0.13$; $p < 0.05$) had significant effects on the farmers' intention to conserve on-farm biodiversity. From a practical viewpoint, these findings provide an argumentation for considering subjective norm, attitude, moral norm, PTIA and knowledge in any planning process to encourage farmers to pursue biodiversity conservation practices.

4. Conclusion

This research adds new knowledge to the existing scientific information about the determinants of intention towards on-farm biodiversity conservation. It presents evidence about effectiveness of the TPB and EMPB models in assessing farmers' intention to preserve biodiversity. The findings illustrated that while the TPB constructs can efficiently explain intention towards biodiversity conservation, incorporating the moral norm, knowledge and PTIA constructs into TPB can increase its predictive power. The study reinforced the available evidence regarding significance of moral norm in development of environmental intention and emphasized the role of knowledge in improving attitude-intention relationships. Overall, the results of the study offered some insights to the policymakers and planners of developing countries for making effective decisions about on-farm biodiversity conservation. Since knowledge can modify the relationship between attitude and

intention, planning and implementing efficient educational programs are emphasized to improve the farmers' knowledge about the concept of biodiversity, its value and the strategies required to protect it at the farm level. In fact, as Ahnström (2009) states, we must institutionalize the conservation of biodiversity on the farmer's mind. In this regard, the contribution of opinion leaders and professional trainers, as well as mass media, is required to clarify the minds of farmers regarding the importance of biodiversity and the destructive effects of intensive agriculture on the natural environment. Moreover, this study suggests that conserving biodiversity on farm requires the programs that can improve social attitude towards biodiversity. In this regard, Deng et al. (2016) perceived that using public education and advertising policies can guide public opinions toward desirable environmental behaviors. In addition, consistent with the theoretical claims and practical findings on the role of threat perception in explaining environmental intentions, the results of the present study confirmed the role of PTIA in guiding the Iranian farmers' intention towards biodiversity conservation. In this regard, the farmers needed to be informed about the negative consequences of intensive agriculture and be provided with the necessary facilities that can help them follow alternative agricultural approaches to increase biodiversity on their farms meantime increasing agricultural production. For example, as EU rural development policy 2007-2013 rewards the farmers who go beyond traditional methods to plant hedgerows, create ponds or leave fields uncultivated (Alony, 2014), the governments of developing countries should provide incentives for the farmers moving toward new and environmental-friendly approaches of weed and pest control. The results also suggested that re-assessment of the policy targets and design is necessary to maximize the farmers' intention to use the management strategies that would balance optimal production of agricultural products with on-farm biodiversity conservation.

In general, this study presents an important insight about effectiveness of TPB in predicting farmers' intention towards biodiversity conservation. In addition, the study provides evidence about incorporation of moral norm, knowledge and PTIA into TPB to increase its power in predicting conservation intention. Combination of various theories such a Norm Activation Model, Protection Motivation Theory or Flow Theory is recommended to future studies in order to identify the other sociological and psychological constructs that might influence farmers' intention towards on-farm biodiversity conservation. In addition, the farmers are just one of the groups that have an intervention to biodiversity conservation at the farm level. It is suggested that, in the future studies, the attitudes and behavioral intentions of the other influential groups, such as agricultural extension agents, policy makers and planners, towards on-farm biodiversity conservation be studied.

Declaration on conflict of interest

Authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gecco.2019.e00698.

References

Ahnström, J., 2009. Farmland Biodiversity – in the Hands and Minds of Farmers Effects of Landscape Structure, Management and the Farmer's Interest in Nature. Dissertation, Department of Ecology, Swedish University of Agricultural Sciences.

Ahnström, J., Höckert, J., Bergea°, H.L., Francis, ChA., Skelton, P., Hallgren, L., 2008. Farmers and nature conservation: what is known about attitudes, context factors and actions affecting conservation? Renew. Agric. Food Syst. 24 (1), 38–47.

Ajzen, I., 1991. The theory of planned behaviour. Organ. Behav. Hum. Decis. Process. 50, 179–211. https://doi.org/10.1016/0749-5978(91)90020-T.

Ajzen, I., Fishbein, M., 1980. Understanding Attitudes and Predicting Social Behavior. Prentice-Hall, Englewood Cliffs, NJ.

Alony, Sh, 2014. Agriculture Sector and Biodiversity Conservation: Best Practice Benchmarking. Workshop of the European Union Business and Biodiversity Platform. http://ec.europa.eu/environment/archives/business/assets/pdf/sectors/FINAL_Agriculture.pdf. (Accessed 15 June 2019).

Aref, A., 2011. Perceived impact of education on poverty reduction in rural areas of Iran. Life Sci. J. 8 (2), 498–501.

Bakhtiari, F., Jacobsen, J.B., Strange, N., Helles, F., 2014. Revealing lay people's perceptions of forest biodiversity value components and their application in valuation method. Glob. Ecol.Conserv. 1, 27–42.

Bamberg, S., Möser, G., 2007. Twenty years after Hines, Hungerford and Tomera: a new meta-analysis of psychosocial determinants of pro-environmental behavior. J. Environ. Psychol. 27, 14–25.

Barson, M., Mewett, J., Paplinska, J., 2012. Trends in on Farm Biodiversity Management in Australia's Agricultural Industries. Caring for Our Country Sustainable Practices Fact Sheet 5. Department of Agriculture, Fisheries and Forestry, Australia.

Bayat, M., Hashemi Hosseini, P., 2010. The introduction of the geo-tourism specialties of Zagros Gorges, Fars province, Iran. In: The 1st International Applied Geological Congress, Mashhad, Department of Geology, Islamic Azad University.

Botetzagias, I., Dima, A.F., Malesios, Ch., 2015. Extending the theory of planned behavior in the context of recycling: the role of moral norms and of demographic predictors. Resour. Conserv. Recycl. 95, 58–67.

Chen, M.F., 2017. Modeling an extended theory of planned behavior model to predict intention to take precautions to avoid consuming food with additives. Food Qual. Prefer. 58, 24–33.

Coromaldi, M., Pallante, G., Savastano, S., 2015. Adoption of modern varieties, farmers' welfare and crop biodiversity: evidence from Uganda. Ecol. Econ. 119, 346–358. https://doi.org/10.1016/j.ecolecon.2015.09.004.

de Leeuw, A., Valois, P., Ajzen, I., Schmidt, P., 2015. Using the theory of planned behavior to identify key beliefs underlying pro-environmental behavior in high-school students: implications for educational interventions. J. Environ. Psychol. 42, 128–138. https://doi.org/10.1016/j.jenvp.2015.03.005.

Deng, J., Sun, P., Zhao, F., Han, X., Yang, G., Feng, Y., 2016. Analysis of the ecological conservation behavior of farmers in payment for ecosystem service programs in eco-environmentally fragile areas using social psychology models. Sci. Total Environ. 550, 382–390. https://doi.org/10.1016/j.scitotenv.2016.01.152.
Dolatkhahi, M., Yousefi, M., BagherNejad, J., Dolatkhahi, A., 2010. Introductory study of the medicinal plant species of Kazeroon, Fars province. J Herb. Drugs 1 (3), 47–56.
Fabrigar, R.L., Petty, E.R., Smith, M.S., Crites, L.S., 2006. Understanding knowledge effects on attitude–behavior consistency: the role of relevance, complexity, and amount of knowledge. J. Personal. Soc. Psychol. 90 (4), 556–577. https://doi.org/10.1037/0022-3514.90.4.556.
Farashi, A., Shariati, M., Hosseini, M., 2017. Identifying biodiversity hotspots for threatened mammal species in Iran. Mamm. Biol. 87, 71–88. https://doi.org/10.1016/j.mambio.2017.06.002.
Fielding, K., McDonald, R., Louis, W.R., 2008. Theory of planned behaviour, identity and intentions to engage in environmental activism. J. Environ. Psychol. 28, 318–326. https://doi.org/10.1016/j.jenvp.2008.03.003.
Firbank, L.G., Petit, S., Smart, S., Blain, A., Fuller, R.J., 2008. Assessing the impacts of agricultural intensification on biodiversity: a British perspective. Phil. Trans. Roy. Soc. 63, 777–787. https://doi.org/10.1098/rstb.2007.2183.
Flaherty, K.L., Turk, P.J., Anderson, J.T., 2019. Comparing stakeholder attitudes toward white-tailed deer and rare plant management in Canaan Valley, West Virginia. Glob. Ecol.Conserv. 17, e00519 https://doi.org/10.1016/j.gecco.2018.e00519.
Gao, L., Wang, Sh., Li, J., Li, H., 2017. Application of the extended theory of planned behavior to understand individual's energy saving behavior in workplaces. Resour. Conserv. Recycl. 127, 107–113.
Greaves, M., Zibarras, L.D., Stride, Ch., 2013. Using the theory of planned behavior to explore environmental behavioral intentions in the workplace. J. Environ. Psychol. 34, 109–120. https://doi.org/10.1016/j.jenvp.2013.02.003.
Greiner, R., 2015. Motivations and attitudes influence farmers' willingness to participate in biodiversity conservation contracts. Agric. Syst. 137, 154–165. https://doi.org/10.1016/j.agsy.2015.04.005.
Guillem, E.E., Murray-Rust, D., Robinson, D.T., Barnes, A., Rounsevell, M.D.A., 2015. Modeling farmer decision-making to anticipate tradeoffs between provisioning ecosystem services and biodiversity. Agric. Syst. 137, 12–23.
Halder, P., Pietarinen, J., Havu-Nuutinen, S., Pöllänen, A., Pelkonen, P., 2016. The theory of planned behavior model and students' intentions to use bioenergy: a cross-cultural perspective. Renew. Energy 89, 627–635.
Herzon, I., Mikk, M., 2007. Farmers' perceptions of biodiversity and their willingness to enhance it through agri-environment schemes: a comparative study from Estonia and Finland. J. Nat. Conserv. 15, 10–25.
Hoeksma, D.L., Gerritzen, M.A., Lokhorst, A.M., Poortvliet, P.M., 2017. An extended theory of planned behavior to predict consumers' willingness to buy mobile slaughter unit meat. Meat Sci. 128, 15–23.
Hojjati, M.H., Boustani, F., 2010. An assessment of groundwater crisis in Iran case study: Fars province. Int.J.Soc.Behav.Educ.Econ.Bus.Ind. Eng. 4 (10), 2066–2070.
Hon Kong Environment Bureau, 2016. Hong Kong Biodiversity Strategy and Action Plan 2016-2021. https://www.afcd.gov.hk/english/conservation/Con_hkbsap/files/HKBSAP_ENG_2. pdf. (Accessed 13 June 2019).
Iranian Department of Environment, 2016. Revised National Biodiversity Strategies and Action Plan (NBSAP2) 2016-2030. https://www.cbd.int/doc/world/ir/ir-nbsap-v2-en.pdf. (Accessed 15 June 2019).
Jackson, L., van Noordwijk, M., Bengtsson, J., Foster, W., Lipper, L., Pulleman, M., Said, M., Snaddon, J., Vodouhe, R., 2010. Biodiversity and agricultural sustainability: from assessment to adaptive management. Curr. Opin.Environ. Sustain. 2, 80–87.
Janmaimool, P., 2017. Application of protection motivation theory to investigate sustainable waste management behaviors. Sustainability 9, 1079. https://doi.org/10.3390/su9071079.
Johnson, R.J., Scicchitano, M.J., 2000. Uncertainty, risk, trust, and information: public perceptions of environmental issues and willingness to take action. Pol. Stud. J. 28, 633–647.
Keshavarz, M., Karami, E., 2014. Farmers' decision-making process under drought. J. Arid Environ. 108, 43–56.
Keshavarz, M., Maleksaeidi, H., Karami, E., 2016. Livelihood vulnerability to drought: a case of rural Iran. Int. J.Disaster Risk.Reduct. 21, 223–230.
Liu, Y., Sheng, H., Mundorf, N., Redding, C., Ye, Y., 2017. Integrating norm activation model and theory of planned behavior to understand sustainable transport behavior: evidence from China. Int. J. Environ. Res. Public Health 14, 1593. https://doi.org/10.3390/ijerph14121593.
Maichum, K., Parichatnon, S., Peng, K., 2016. Application of the extended theory of planned behavior model to investigate purchase intention of green products among Thai consumers. Sustainability 8, 1077.
Maleksaeidi, H., Karami, E., Zamani, Gh, Rezaei-Moghaddam, K., Hayati, D., Masoudi, M., 2016. Discovering and characterizing farm households' resilience under water scarcity. Environ. Dev. Sustain. 18, 499–525.
Malek-Saeidi, H., Rezaei-Moghaddam, K., Ajili, A., 2012. Professionals' attitudes towards organic farming: the case of Iran. J. Agric. Sci. Technol. A 14, 37–50.
Marquit, J.D., 2008. Threat Perception as a Determinant of Pro-environmental Behaviors: Public Involvement in Air Pollution Abatement in Cache Valley, Utah. Department of Psychology, Utah State University, United States. MSc thesis.
McFarlane, B.L., Boxall, P.C., 2003. The role of social psychological and social structural variables in environmental activism: an example of the forest sector. J. Environ. Psychol. 23, 79–87.
Meijer, S.S., Catacutan, D., Sileshi, G., Nieuwenhuis, M., 2015. Tree planting by smallholder farmers in Malawi: using the theory of planned behaviour to examine the relationship between attitudes and behavior. J. Environ. Psychol. 43, 1–12.
Meyers, L.S., Gamst, G., Guarino, A.J., 2006. Applied Multivariate Research: Design and Interpretation. Sage, London.
Mullan, B.A., Wong, C., Kothe, E.J., 2013. Predicting adolescents' safe food handling using an extended theory of planned behavior. Food Control 31, 454–460.
Nafarzadegan, A.R., Rezaeian Zadeh, M., Kherad, M., Ahani, H., Gharehkhani, A., Karampoor, M.A., Kousari, M.R., 2012. Drought area monitoring during the past three decades in Fars province, Iran. Quat. Int. 250, 27–36.
Niankara, I., Zoungrana, D.T., 2018. Interest in the biosphere and students environmental awareness and optimism: a global perspective. Glob. Ecol.Conserv. 16, e00489.
Park, K.J., 2015. Mitigating the impacts of agriculture on biodiversity: bats and their potential role as bio-indicators. Mamm. Biol. 80, 191–204.
Pieniak, Z., Aertsens, J., Verbeke, W., 2010. Subjective and objective knowledge as determinants of organic vegetables consumption. Food Qual. Prefer. 21, 581–588.
Poškus, M.S., 2015. Predicting recycling behavior by including moral norms into the theory of planned behavior. PSICHOLOGIJA 52, 22–32.
Prochaska, J.O., Di Clemente, C.C., 1982. Transtheoretical therapy: toward a more integrative model of change. Psychother. Theory Res. Pract. 19 (3), 276–288.
Raymond, L., Schneider, L.U., 2014. Personal moral norms and attitudes toward endangered species policies on private land. Conserv. Soc. 12 (1), 1–15.
Rosenstock, I.M., 1974. Historical origins of the health belief model. Health Educ. Monogr. 2 (4), 328–335.
Scheaffer, R.L., Mendenhall, W., Ott, L., 1979. Elementary Survey Sampling. Duxbuy Press, Belmont.
Scherr, S.J., McNeely, J.A., 2008. Biodiversity conservation and agricultural sustainability: towards a new paradigm of 'ecoagriculture' landscapes. Philos. Trans. R. Soc. Lond. Ser. B Biol. Sci. 363 (1491), 477–494.
Schwartz, S.H., 1977. Normative influences on altruism. Adv. Exp. Soc. Psychol. 10, 222–280.
Shin, Y.H., Hancer, M., 2016. The role of attitude, subjective norm, perceived behavioral control, and moral norm in the intention to purchase local food products. J. Foodserv. Bus. Res. 19 (4), 338–351. https://doi.org/10.1080/15378020.2016.1181506.
Statistical Center of Iran, 2014. Statistical Yearbook 2013-2014. http://www.amar.org.ir/Portals/0Files/fulltext/1393/n_tsokkk_93-v3.pdf. (Accessed 29 April 2017).
Stern, P.C., 1992. Psychological dimensions of global environmental change. Annu. Rev. Psychol. 43, 269–302.

The Fifth National Report to the convention on Biological Diversity, 2015. https://www.cbd.int/doc/world/ir/ir-nr-05-en.pdf. (Accessed 19 April 2017).
The Forth National Report to the convention on Biological Diversity, 2010. https://www.cbd.int/doc/world/ir/ir-nr-04-en.pdf. (Accessed 19 April 2017).
United Nations Development Programme, 2016. Iran to Expand Successful 10-Year-Old UNDP Zagros Project with Own Funds. http://www.ir.undp.org/content/iran/en/home/presscenter/articles/2016/02/15/iran-to-expand-successful-10-year-old-undp-zagros-project-with-own-funds.html. (Accessed 19 April 2017).
Van Thanh, N., Yapwattanaphun, Ch., 2015. Banana farmers' adoption of sustainable agriculture practices in the Vietnam Uplands: the case of Quang Tri Province. Agric. Sci.Procedia 5, 67–74. https://doi.org/10.1016/j.aaspro.2015.08.010.
Vinodh, S., Joy, D., 2012. Structural equation modeling of sustainable manufacturing practices. Clean Technol. Environ. Policy 14, 79–84.
Wu, W.W., 2010. Linking Bayesian networks and PLS path modeling for causal analysis. Expert Syst. Appl. 37, 134–139.
Yazdanpanah, M., Forouzani, M., 2015. Application of the theory of planned behaviour to predict Iranian students' intention to purchase organic food. J. Clean. Prod. 107, 342–352. https://doi.org/10.1016/j.jclepro.2015.02.071.
Zhang, X., Geng, G., Sun, P., 2017. Determinants and implications of citizens' environmental complaint in China: integrating theory of planned behavior and norm activation model. J. Clean. Prod. 166, 148–156.